Types of Variables & Interface

05 July 2026 09:03

Today will cover:

1 Local variable	2 Instance variable	3 Static variable	4 Static block
5 Access modifiers	6 Non-access modifiers	7 Final	8 Abstract class
9 Interface	10 String	11 StringBuffer	12 StringBuilder

Q If we create 100 student objects, should every student have separate name ?

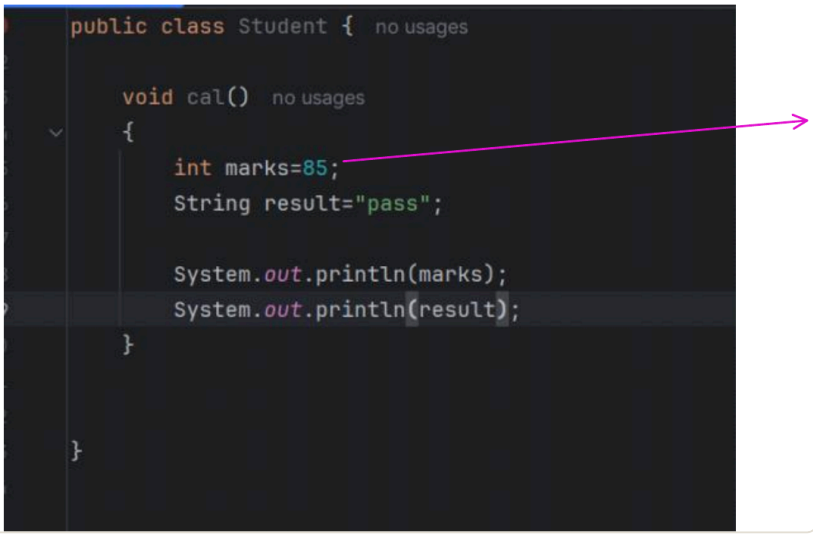
A Yes

Q If all students belong to the same institute, should every object store a separate institute name ?

A Institute will be same for all students : NO

Types of variables (area) in java

01 Local variable



Marks and result are local variables

They exist only inside `cal()`

Important points:

1	Local variables are created when method/block starts
2	Local variables are destroyed when method/block ends
3	Local variables cannot be used outside their method/block
4	Local variables do not get default values
5	We must initialize local variables before using them

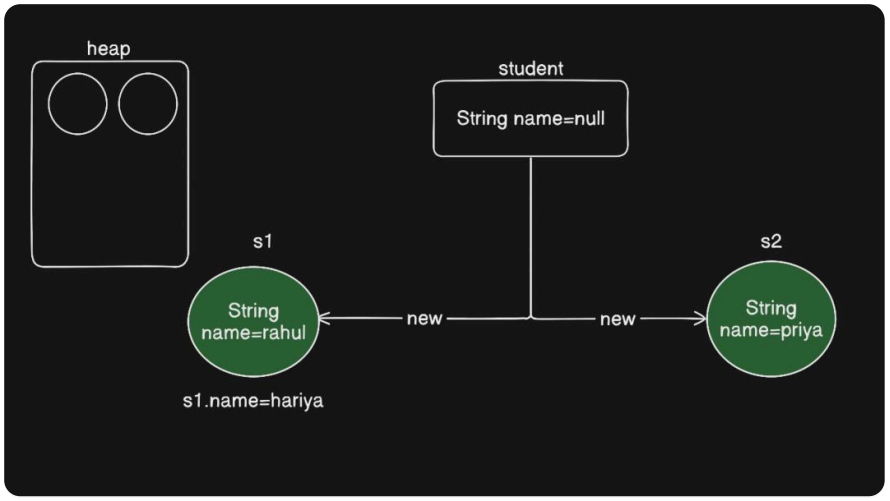
02 Instance variable:

An instance variable is declared inside a class but outside any method, constructor or block:

Instance variable = object-level variable

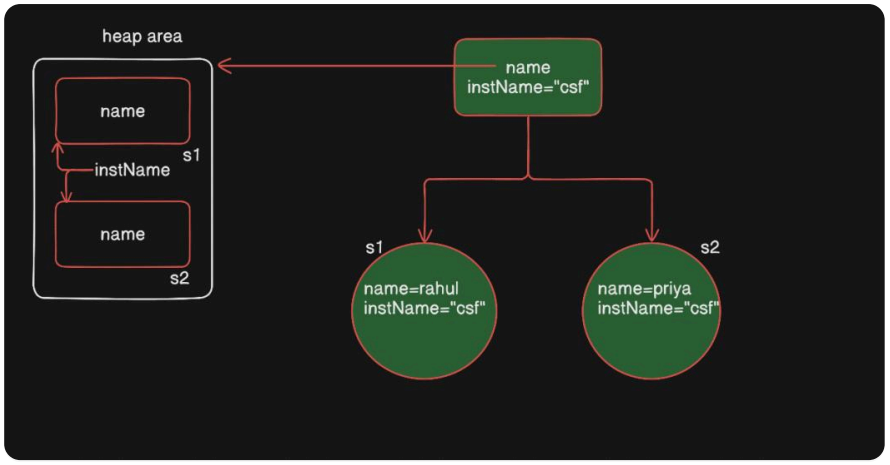
```
public class Student {  no usages
    String name;  no usages
    int age;  no usages
    int marks;  no usages

}
```



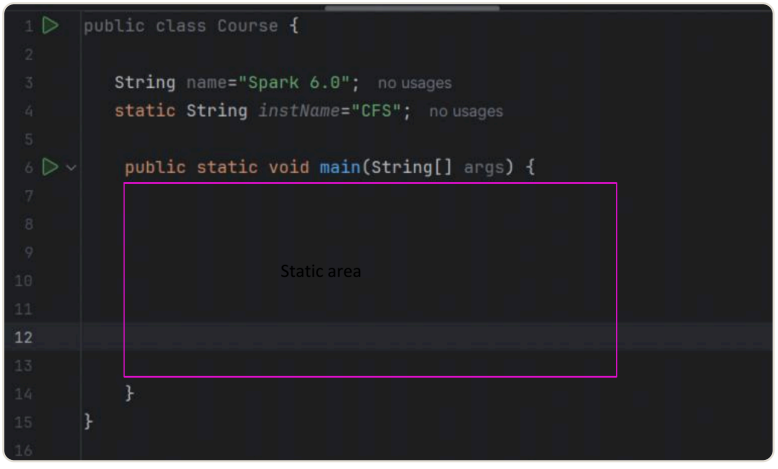
03 Static variable:

Static variable = class-level variable shared by all objects



Instance variable Vs Static variable

Point	Instance variable	Static variable
Belongs to	Object	Class
Memory copy	Separate copy for each object	Single copy
Access	Reference variable	Class name



Static area

In static area we can only use static variables

Non static area

We can use both instance and static variables

Q1 **Why main method is static ?**

JVM can call main method without creating object of the class.

04 Static block

A **static block** is used to initialize static data.

05 Access Modifiers

Access modifiers control visibility.

Java has four access levels:

1. `private`
2. `default`
3. `protected`
4. `public`

Important: If we do not write any access modifier, it is default access.

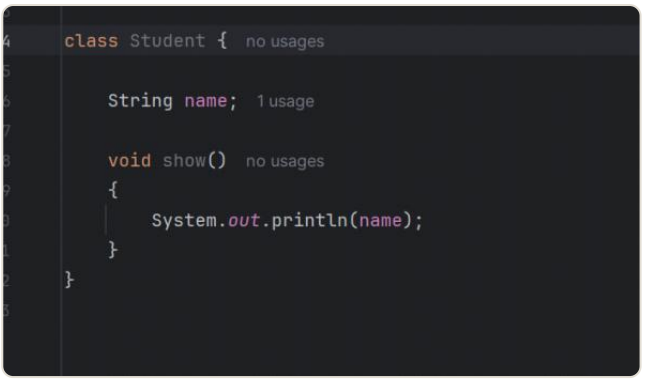
Modifier	Same Class	Same Package	Child class Different package	Anywhere
`private`	Yes	No	No	No
`default`	Yes	Yes (child class/other class)	No	No
`protected`	Yes	Yes	Yes	No
`public`	Yes	Yes	Yes	Yes

Private:

Private variables and methods will not be available for inheritance and can only be accessed inside the class.

Default:

Default member can be accessed inside the same package



Student class, name variable and show method have default access.

> Java does not allow `protected` and `private` on top-level classes.

06 Non-Access modifiers

1. **Static : belongs to class**
2. **Final :**
3. **Abstract :**

07 Final:

> **Final** means cannot be changed.

It can be used with

`1. Variable` `2. Method` `3. Class`

Q **Difference between final , finally, finalize()**

`final`	`finally`	`finalize()`
keyword	block	--> G.C

Real example: final

Math formula : pi	Tax percentage

08 Abstract class:

> An abstract class is a class declared using `abstract` keyword.

SIMPLE DEFINITION:

Abstract class : incomplete class that can contain abstract and normal methods.

> **Note:**
>
> 1. If any class has abstract method then we must use abstract keyword with the class.
> 2. We cannot create object of abstract class directly.

09 Interface: (1.7V)

> **Interface** = contract that says what a class must do

? **Why abstract class is not 100% abstraction and why interface is 100% abstraction ?**